

Physics 214/224, Quiz 3

Primary instructor: Yurii Maravin

03/29/2019

For grading purposes

Problem 1	
Problem 2	
Problem 3	
Problem 4	
Studio	
Total:	

Key

WID:_____

NAME:_____

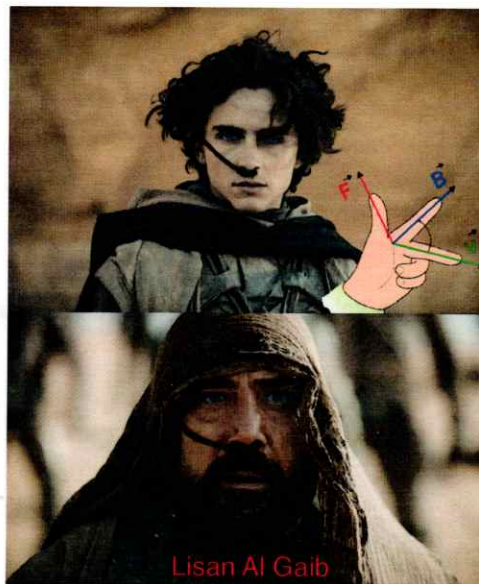
Instructions. Print and sign your name on this quiz and on your scantron card. In doing so, you are acknowledging the KSU Honor Code: "On my honor as a student I have neither given nor received unauthorized aid on this assignment."

For two quick points, circle your studio instructor:

Lohman (TU 7:30) Maravin (TU 9:30) Maravin (TU 11:30)

Han (WF 9:30)

Work alone and answer all questions. Part I questions must be answered on the Scantron cards. Put your name on your card. Color in the correct box completely for every answer with a pencil. If you make a mistake, erase thoroughly. **Don't forget to color in the boxes for your WID. If we have to correct this by hand we may take off five points from your score!** Part II must be answered in the space provided on the test. The last page is a detachable equation sheet. You may use a calculator. You may ask the proctors questions of clarification. Show your work in a clear and organized manner. You may receive partial credit for partial solutions. Solutions lacking supporting calculations will receive no points.



Part I: All questions to be answered on your Scantron card (48 points total)

1. (3 points) What is the unit for magnetic field

- (a) Coulomb
- (b) Ampere
- ☒ (c) Tesla
- (d) Dollar

2. (3 points) Which hand should **NOT** be used for hand rules discussed in EP2 class

- (a) Right
- ☒ (b) Left

3. (3 points) How are magnet poles called?

- (a) Sweet and Sour
- (b) Happy and Sad
- ☒ (c) North and South
- (d) Top and Bottom

4. (3 points) Your EP2 professors name is

- (a) Donald Trump
- (b) Willy Wonka
- (c) Josef Jugashvili (also known as Josef Stalin)
- ☒ (d) Yurii Maravin

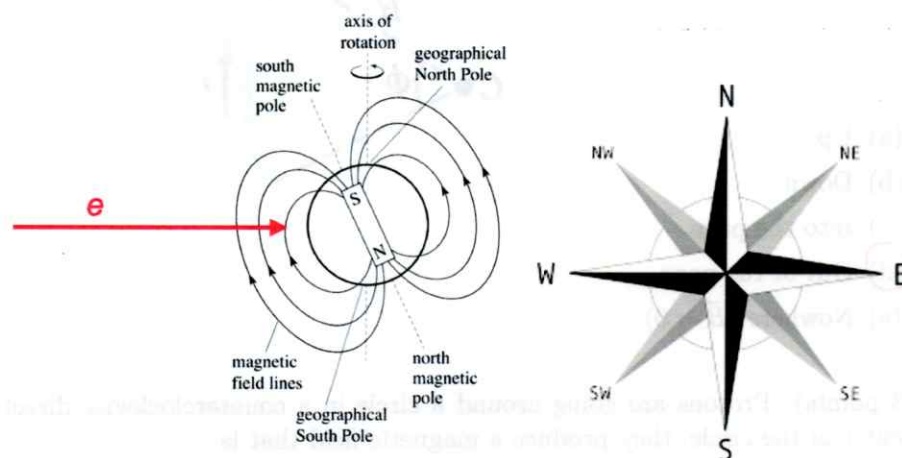
5. (3 points) A charged particle with a velocity $\vec{v}$, **perpendicular** to the magnetic field $\vec{B}$ will travel in a

- (a) straight line
- (b) ellipse
- (c) rectangular path
- ☒ (d) circle
- (e) McLaren 720S

6. (3 points) A charged particle with a velocity $\vec{v}$, **parallel** to the magnetic field $\vec{B}$ will travel in a

- ☒ (a) straight line
- (b) ellipses
- (c) rectangular path
- (d) circle
- (e) Bugatti Chiron Profilée

7. (3 points) Suppose an electron e is headed at us from outer space in the direction shown in the figure below. Which way will the Earth's magnetic field deflect it? Assume the Earth's magnetic field points due north. The relative location of north, east, south, and west, as viewed from the top, is provided in case it is needed.

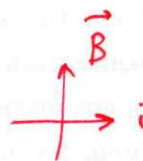


- (a) North
(b) South
(c) East
(d) West
(e) Nowhere
8. (3 points) Which current-carrying wire configuration would produce a magnetic field pattern most similar to that produced by the earth?
- (a) A long straight wire
(b) A toroid
(c) A solenoid
(d) A current sheet

The next three questions are based on the following situation: a power line carries 0.1 A of current in a west-to-east direction across Kansas.

9. (3 points) Which way would the magnetic force from the Earth's magnetic field act on this power line?

- (a) Up
(b) Down
(c) North
(d) South
(e) Nowhere

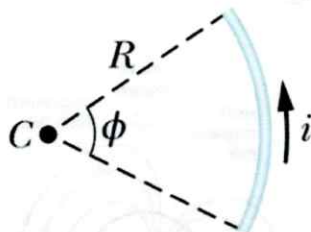


10. (3 points) Estimate the magnitude of the magnetic force on a 100m length of the power line.

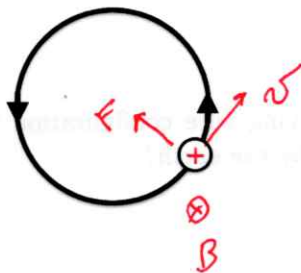
- (a) 5×10^{-4} N
(b) 5×10^{-1} N
(c) 5×10^2 N
(d) 5×10^5 N
(e) 5×10^6 pineapples

$$0.1 \text{ A} \times 100 \text{ m} \times 5 \cdot 10^{-5} \text{ T} = 5 \times 10^{-4} \text{ N}$$

11. (3 points) A segment of a wire carries a current i as indicated in the figure below. Which way will the magnetic field from this current be pointing at C?



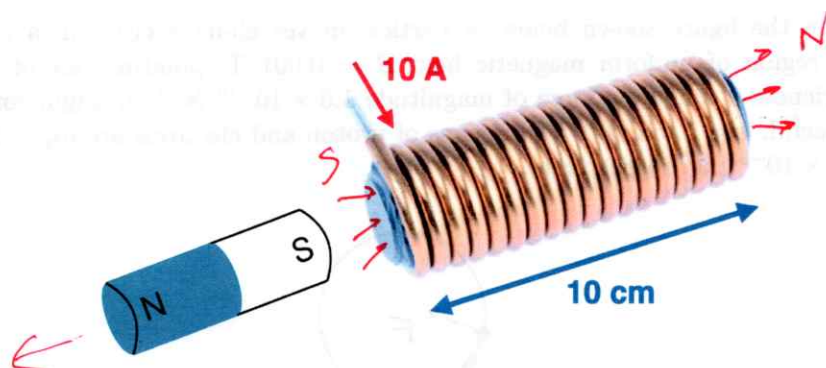
- (a) Up
 - (b) Down
 - (c) Into the page
 - ☒ (d) Out of the page
 - (e) Nowhere ($\vec{B} = 0$)
12. (3 points) Protons are going around a circle in a counterclockwise direction as shown. At the center of the circle, they produce a magnetic field that is:



$$\vec{F} = q\vec{v} \times \vec{B}$$

- ☒ (a) into the page
 - (b) out of the page
 - (c) to the left
 - (d) to the right
 - (e) zero
13. (3 points) What would you get if you took a bar magnet and cut it in half?
- (a) two pieces, one a north pole, and one a south pole
 - ☒ (b) two shorter magnets, each with a north pole and a south pole
 - (c) two pieces, both non-magnetic
 - (d) a molten mess from the energy released from within the magnet upon cutting
 - (e) an electric shock

The next two problems refer to the solenoid described below, with the current flowing in the direction shown.



14. (3 points) The magnetic field at the center of the solenoid has a magnitude

- (a) $B = 1.0 \times 10^{-4} \text{ T}$
- (b) $B = 2.3 \times 10^{-3} \text{ T}$
- (c) $B = 1.9 \times 10^{-3} \text{ T}$
- (d) $B = 3.8 \times 10^{-2} \text{ T}$

$$4\pi \cdot 10^{-7} \frac{18}{0.1} \cdot 101 \approx 2.3 \cdot 10^{-3} \text{ T}$$

15. (3 points) The bar magnet to the left of the solenoid in the picture above will be

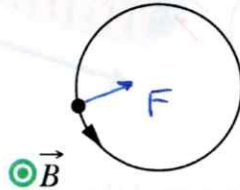
- (a) Pushed to the right
- (b) Pushed to the left
- (c) Stay the way it is
- (d) Start rotating clockwise
- (e) Start rotating counterclockwise

16. (3 points) A piece of unknown material develops a small magnetic field pointing toward the north pole of the bar magnet when the bar magnet is brought close. The field goes away when the bar magnet is removed. The unknown material is

- (a) non-magnetic
- (b) diamagnetic
- (c) paramagnetic
- (d) ferromagnetic

Part II: Work out this part in the space provided. Show your work!
(50 points total)

17. (14 points) In the figure shown below, a particle moves along a circle in a counterclockwise fashion, in a region of uniform magnetic field $B = 0.001$ T, pointing out of the page. The particle experiences a magnetic force of magnitude 1.6×10^{-15} N. You might find the following information useful: $e = 1.6 \times 10^{-19}$ C, masses of proton and electrons are $m_p = 1.67 \times 10^{-27}$ kg and $m_e = 9.1 \times 10^{-31}$ kg, respectively.



- (a) (3 points) Is the particle a proton or an electron? (Reminder: as for all write-up questions, you have to explain your choice to get the full credit)

(+1) From RHR the particle must be an electron (+1)
 $\vec{F} = q\vec{v} \times \vec{B}$ (+1)

Explanation +2
 Correct answer +1

- (b) (3 points) Determine the particle's speed.

$$F = qvB \Rightarrow v = \frac{F}{qB} = \frac{1.6 \cdot 10^{-15} \text{ N}}{1.6 \cdot 10^{-19} \text{ C} \cdot 10^{-3} \text{ T}} = 10^7 \text{ m/s}$$

(+1) (+2)

- (c) (5 points) Determine the radius of the circle.

$$R = \frac{mv}{qB} = \frac{9.1 \cdot 10^{-31} \text{ kg} \cdot 10^7 \text{ m/s}}{1.6 \cdot 10^{-19} \text{ C} \cdot 10^{-3} \text{ T}} \approx 0.057 \text{ m}$$

(+3) (+2)

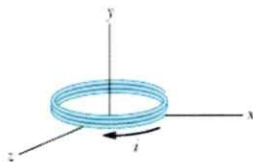
Do not remove points if they made a mistake in (a) and use proton mass

- (d) (3 points) Determine the period of the motion.

$$T = \frac{2\pi R}{v} = \frac{2\pi \cdot 0.057 \text{ m}}{10^7 \text{ m/s}} = 3.58 \cdot 10^{-8} \text{ s} \approx 35.8 \text{ ns}$$

(+2) (+1)

18. (14 points) A coil shown below carries current $i = 2.00$ A in the direction indicated, is parallel to an xz plane, has 3 turns and an area of 4.00×10^{-3} m², and lies in a uniform magnetic field $\vec{B} = (1.00\hat{i} + 2.00\hat{j} + 4.00\hat{k})$ mT.



- (a) (4 points) What is the magnitude of the magnetic dipole of the coil?

$$\mu = iA = 3 \cdot 200 \text{ A} \cdot 4 \cdot 10^{-3} \text{ m}^2 = 2.4 \cdot 10^{-2} \text{ A m}^2$$

- (b) (2 points) What is the direction of the magnetic dipole moment (express your answer in terms of axes unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$)

Using RHR the direction is in the negative $\hat{j}$ direction (opposite y-axis direction): $-\hat{j}$ Explanation (+)

Explanation (+1)
Correct answer (+1)

or $\vec{\mu} = -2.4 \cdot 10^{-2} \text{ A} \cdot \text{m}^2 \hat{j}$

- (c) (4 points) What is the potential energy of this magnetic dipole?

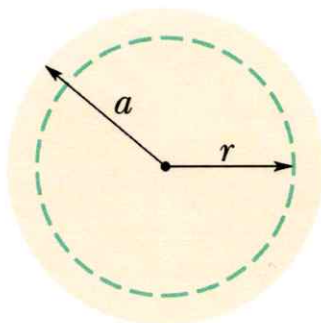
$$U = -\vec{\mu} \cdot \vec{B} = -(1.00\hat{i} + 2\hat{j} + 4\hat{k})(-2.4 \cdot 10^{-2} \text{ A} \cdot \text{m}^2 \hat{j}) \cdot 10^{-3} \text{ T}$$

$$= 2 \cdot 2.4 \cdot 10^{-2} \text{ A} \cdot \text{m}^2 \text{ T} \cdot 10^{-3} = 4.8 \cdot 10^{-5} \text{ J}$$

- (d) (4 points) What is the torque on this magnetic dipole moment due to the external magnetic field? (*Hint: you might want to remember that $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$, and that for any vectors $\vec{A}$ and $\vec{B}$: $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$*)

$$\begin{aligned}\vec{\tau} &= \vec{\mu} \times \vec{B} = (-24 \cdot 10^{-2} \text{ A m}^2 \hat{j}) \times (\hat{i} + 2\hat{j} + 4\hat{k}) \cdot 10^{-3} \text{ T} = \\ &= (-24 \cdot 10^{-5} \text{ J}) \hat{j} \times \hat{i} + (-4 \cdot 24 \cdot 10^{-5} \text{ J}) (\hat{j} \times \hat{k}) = \\ &= (2.4 \cdot 10^{-5} \text{ J}) \hat{k} - (9.6 \cdot 10^{-5} \text{ J}) \hat{i}\end{aligned}$$

19. (12 points) The figure below portrays the cross-section of a long cylindrical conducting cylinder of radius $a = 1.00$ m. The current in the cylinder is 10 A and uniformly distributed.



- (a) (3 points) What is the magnetic field due to the current at $r = 0$ m?

Due to symmetry B must be zero
 Explanation (+2) (+1)

Also ok if they use
 Ampère law and
 take a limit:
 $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 i_{enc}$
 $= 2\pi r B = \mu_0 j \cdot \pi r^2$
 ↑
 density

- (b) (3 points) What is the current density of the wire?

$$j = \frac{10A}{\pi r^2} = \frac{10A}{\pi \cdot (1m)^2} \approx 3.18 \frac{A}{m^2}$$

(+2) (+1)

$$B = \frac{\mu_0 j r}{2}$$

$\lim_{r \rightarrow 0} B = 0.$

- (c) (6 points) What is the magnetic field due to the current at $r = 0.7$ m?

(+1) Use Ampère's law $\oint \vec{B} d\vec{\ell} = \mu_0 i_{enc}$

(+1) $2\pi r B = \mu_0 \cdot i_{enc}$

(+2) $i_{enc} = \pi r^2 \cdot j \Rightarrow$

$$B = \frac{\mu_0 \cdot \pi r^2 j}{2\pi r} = \frac{\mu_0 r \cdot j}{2} = \frac{4\pi \cdot 10^{-7} \frac{Tm}{A} \cdot 0.7m \cdot 3.18 \frac{A}{m^2}}{2}$$

$$= 1.4 \cdot 10^{-6} T$$

(+2)

20. (10 points) Suppose we wish to magnetize a huge volume of a cylinder-shaped spectrometer at the CERN Large Hadron Collider to a value of 4.0 Tesla. We choose a solenoid to achieve this. The solenoid is 20 meters long and 6 meters in diameter. A Physicists-R-Us power supply can deliver 20,000 A to the solenoid.

- (a) (4 points) Show that we need to wrap a conductor outside of the cylinder a total of 3183 times over its full length to produce this field with a given current.

$$B = \underbrace{\mu_0 n i}_{(+2)} = \mu_0 \underbrace{\frac{N}{L} i}_{(+2)} \Rightarrow N = \frac{BL}{\mu_0 i} = \frac{4T \cdot 20m}{4\pi \cdot 10^{-7} \frac{Tm}{A} \cdot 20,000A} \approx 3183$$

- (b) (2 points) Show that we would need about 60 km of wire

$$L = N \cdot \underbrace{\text{circumference}}_{(+2)} = 3183 \cdot \pi \cdot 6m \approx 60,000m$$

- (c) (4 points) Suppose the wire had a resistance of 1 Ω /km. How much power would be required to operate the solenoid in this case? Does the value you get seem practically achievable? For reference, the Wolf Creek nuclear plant in Burlington, KS can produce electrical power of 1.25 GW. (+1)

$$P = i^2 R = (20,000A)^2 \cdot \frac{1\Omega}{km} \cdot 60km \approx 2.4 \cdot 10^{10} W$$

$$P = 24 \cdot 10^9 W = 24 GW \leftarrow (+1)$$

explanation (+2) { This is much bigger than what a regular nuclear plant can produce $\rightarrow$ not achievable